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Assignment 2
Due on Feb 25, 2020

One researcher found one organism has a very interesting property. After many steps' purification, he was able to get a small amount of protein. Then he used Mass spectrometry (MS) to find out the sequence and the molecular weight of the protein. The protein sequence is provided in the form of single-letter amino acid code.

1 medaknikkg papfypledg tageqlhkam kryalvpgti aftdahievd ityaeyfems
61 vrlaeamkry glntnhrivv csenslqffm pvlgalfigv avapandiyn erellnsmgi
121 sqptvvfvsk kglqkilnvq kklpiiikii imdsktdyqg fqsmtyfvts hlppgfneyd
181 fvpesfdrdk tialimnssg stglpkgval phrtacvrfs hardpifgnq iipdtailsv
241 vpfhhgfgmf ttglylicgf rrvlmyrfee elfrlsqdy kigsallvpt lfsffakstl
301 idkydlsnlh eiasggapls kevgeavakr fhlpgirqgy gltettsail itpegddkpg
361 avgkvvpffe akvvdldtgk tlgnvqrgel cvrgpmimsg yvnnpeatna lidkdgwllhs
421 gdiaywdede hffivdrks likykyqva paelesillq hpnifdagva glpdddagel
481 paavvlehg ktmtekeivd yvasqvtak klrggvfvvd evpkglgtgkl darkireili
541 kakkggkiav

551

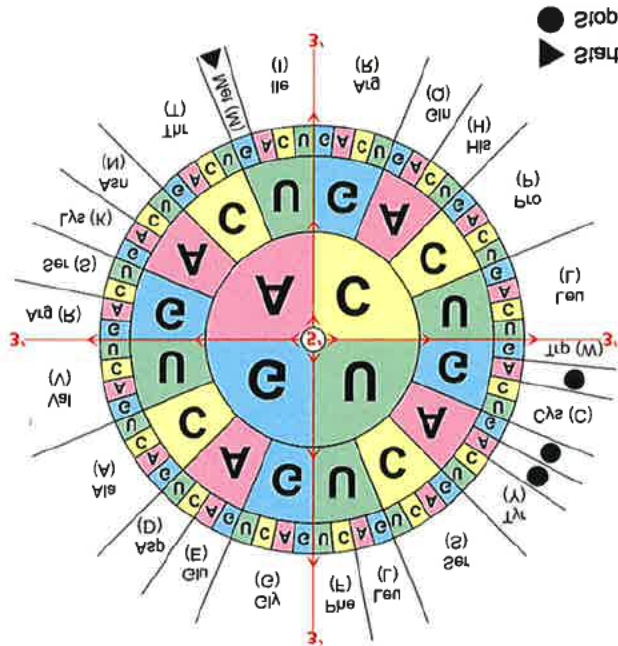
He needs to make more protein to further characterize it. Unfortunately, he does not have enough organisms to provide him enough protein for the characterization.

You need to help him by doing the following things:

(a) Propose a method to make more protein (as detailed as possible) ?

First, we need to find the most likely codon ordering method based on the amino acid sequence of the protein. According to the codon table shown below:

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Then find the arrangement of tRNA based on the amino acid codon sequence found. The codon table was also used to find the tRNA sequence. Then find the sequence of mRNA according to tRNA. This can be reversed to find the DNA sequence for transcription. Finding the DNA sequence to be transcribed and then carrying out the DNA replication process produces a large amount of DNA possessing this protein. Then the DNA transcription and translation process can get a lot of protein that he want.

(b) Tell him the molecular weight of this protein

The protein weighs 60.65 kilodaltons.

(c) Convert the protein sequence to corresponding DNA sequence (Please specific which codon you are using)

The most likely DNA sequence is shown below:

atggaagatgcgaaaaacattaaaaaaggccccggcgccgtttatccgctggaagatggcaccgcgggcgaacagctgcataaagc
gatgaaacgctatgcgctggcgccgggcaccattgcgtttaccgatgcgcattgaagtggatattacctatgcggaatatttgaata
gcgtgcgcctggcggaagcgatgaaacgctatggcctgaacaccaaccatgcattgtggtgtgcagcgaaaaacagcctgcagttt
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aaaattattattatggatagcaaaaccgattatcagggtttcagagcatgtatacctttgtgaccagccatctgccgcccggcgttaacg
aatatgattttgtgccgaaagctttgatcgcgataaaaccattgcgctgattatgaacagcagcggcagcaccggcctgccgaaagg
cgtggcgctgccgatcgacccgctgcgtgcgctttagccatgcgcgcatccgattttggcaaccagattattccggataccgcg
attctgagcgtggtgccgtttcatcatggctttggcatgtttaccaccctgggctatctgatttgcggctttcgcgtggtgctgatgtatgc
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gatgataaaccggcgcggtgggcaaaagtgggtgccgtttttgaagcgaagtggtggatctggataccggcaaaaccctgggcgtg
aaccagcggcggaactgtgcgtgcgcggcccgatgattatgagcggctatgtgaacaaccgggaagcgaccaacgcgctgattg
ataaagatggctggctgcatagcggcgatattgcgtattgggatgaagatgaacattttttattgtgatcgctgaaaagcctgattaa
atataaaggctatcaggtggcgccggcggaactggaaagcattctgctgcagcatccgaacattttgatcgggcggtggcgccct
cccggatgatgatgcggcggaactgccggcgccgggtggtggtgctggaacatggcaaaaccatgaccgaaaaagaaattgtggatt
atgtggcgagccaggtgaccaccgcgaaaaaactgcgcggcgccgctggtgtttgttgatgaagtgcggaaggcctgaccggcaa
actggatgcgcgcaaaattcgcgaaattctgattaaagcgaaaaaaggcggcaaaattgcggtg

The possibilities for each amino acid codon are shown below:

AmAcid	Codon	Number	/1000	Fraction	AmAcid	Codon	Number	/1000	Fraction
Gly	GGG	50527.00	11.12	0.15	Trp	TGG	65630.00	14.45	1.00
Gly	GGA	39036.00	8.59	0.12	End	TGA	4428.00	0.97	0.30
Gly	GGT	114185.00	25.14	0.34	Cys	TGT	23461.00	5.17	0.45
Gly	GGC	130043.00	28.63	0.39	Cys	TGC	28747.00	6.33	0.55
Glu	GAG	83804.00	18.45	0.32	End	TAG	1172.00	0.26	0.08
Glu	GAA	179460.00	39.51	0.68	End	TAA	9006.00	1.98	0.62
Asp	GAT	146794.00	32.32	0.63	Tyr	TAT	75774.00	16.68	0.58
Asp	GAC	87759.00	19.32	0.37	Tyr	TAC	55847.00	12.30	0.42
Val	GTG	115687.00	25.47	0.36	Leu	TTG	60322.00	13.28	0.13
Val	GTA	51020.00	11.23	0.16	Leu	TTA	62823.00	13.83	0.13

Val	GTT	86572.00	19.06	0.27	Phe	TTT	100128.00	22.05	0.57
Val	GTC	67356.00	14.83	0.21	Phe	TTC	74885.00	16.49	0.43
Ala	GCG	146264.00	32.20	0.34	Ser	TCG	39546.00	8.71	0.15
Ala	GCA	93390.00	20.56	0.22	Ser	TCA	35837.00	7.89	0.13
Ala	GCT	73677.00	16.22	0.17	Ser	TCT	42367.00	9.33	0.16
Ala	GCC	113412.00	24.97	0.27	Ser	TCC	40365.00	8.89	0.15
Arg	AGG	7423.00	1.63	0.03	Arg	CGG	25751.00	5.67	0.10
Arg	AGA	12345.00	2.72	0.05	Arg	CGA	16607.00	3.66	0.07
Ser	AGT	41544.00	9.15	0.15	Arg	CGT	93997.00	20.70	0.37
Ser	AGC	70867.00	15.60	0.26	Arg	CGC	96053.00	21.15	0.38
Lys	AAG	51685.00	11.38	0.25	Gln	CAG	130898.00	28.82	0.66
Lys	AAA	156169.00	34.38	0.75	Gln	CAA	67129.00	14.78	0.34
Asn	AAT	84846.00	18.68	0.46	His	CAT	57585.00	12.68	0.57
Asn	AAC	98018.00	21.58	0.54	His	CAC	43743.00	9.63	0.43
Met	ATG	123604.00	27.21	1.00	Leu	CTG	231373.00	50.94	0.49
Ile	ATA	24233.00	5.34	0.09	Leu	CTA	18067.00	3.98	0.04
Ile	ATT	135873.00	29.92	0.50	Leu	CTT	51442.00	11.33	0.11
Ile	ATC	111878.00	24.63	0.41	Leu	CTC	48147.00	10.60	0.10
Thr	ACG	63696.00	14.02	0.26	Pro	CCG	101467.00	22.34	0.51
Thr	ACA	35995.00	7.93	0.15	Pro	CCA	38663.00	8.51	0.20
Thr	ACT	43256.00	9.52	0.18	Pro	CCT	32678.00	7.19	0.17

Thr ACC 103121.00 22.70 0.42

Pro CCC 24383.00 5.37 0.12

(d) What is the putative function of this protein? (using protein blast tools in one of the provided bioinformatics databases)

This protein is firefly luciferase-polyprotein fusion protein [synthetic construct].

Sequence ID: AFE85520.1 Length: 1889

Firefly luciferase has made extensive use of reports in molecular biology and bioanalysis systems due to its high quantum size of bioluminescence, complementation of stable mutant forms of enzymes with prescribed spectral characteristics, and abundant alternative bacterial expression systems that express recombinant proteins molecule.

Luciferase is a heat-sensitive protein used in protein denaturation studies to test the protective ability of heat shock proteins. Opportunities for using luciferase continue to expand. In a luciferase reaction, luciferase emits light when it acts on an appropriate luciferin substrate. Photon emission can be detected by a photosensitive device such as a light meter or a modified optical microscope. This allows observation of biological processes. Since luciferase bioluminescence does not require photoexcitation, the autofluorescence is extremely small, so there is virtually no background fluorescence.

(e) Give at least three applications of this protein

1. System where luciferase attaches to the surface of a target and detects a bioluminescent signal from the specific complex oxide formed.

2. In biological research, luciferase is often used as a reporter gene to assess transcriptional activity in cells transfected with a gene construct containing a luciferase gene under the control of a target promoter. In addition, fluorescent molecules that are converted to luciferin under the activity of a specific enzyme can be used for enzyme activity detection in conjugation or two-step luciferase detection. Such substrates have been used to detect caspase activity and cytochrome P450 activity.

3. Luciferase can be used to detect cellular ATP levels in cell viability assays or kinase activity assays. Luciferase can act as an ATP sensor protein through biotinylation. By binding to the streptavidin-biotin complex, biotinylation fixes luciferase to the cell surface. Luciferase can detect the outflow of ATP in cells and effectively display the real-time release of ATP through bioluminescence. In addition, increasing the luminous intensity by changing certain amino acid residues in the protein sequence can make luciferase more sensitive to ATP detection.